1. Literature review – Publication 1

	Verified information
	Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records.
	Strack, B., DeShazo, J.P., Gennings, C., Olmo, J.L., Ventura, S., Cios, K.J. and Clore, J.N. (2014) 'Impact of HbA1c measurement on hospital readm 70,000 clinical database patient records', BioMed Research International, 2014, Article ID 781670. https://doi.org/10.1155/2014/781670.
	https://pmc.ncbi.nlm.nih.gov/articles/PMC3996476/
	Health Facts electronic health record database. The study extracted 101,766 qualifying inpatient diabetic encounters from 130 US hospitals and integ (1999–2008). The final analytical sample was reduced to 69,984 encounters after exclusions and selection of one encounter per patient.
	Multivariable logistic regression to examine the relationship between HbA1c measurement and early readmission while controlling for demographic, d admission-related covariates.
	The authors applied eligibility criteria to identify inpatient diabetic encounters with a length of stay of 1–14 days, laboratory testing and medication adr with substantial missingness or limited relevance were removed or handled as missing categories. Multiple encounters per patient were addressed by encounter for the primary analysis. Hospice discharges and deaths were excluded.
	The study reported that HbA1c testing was performed infrequently (18.4%) and that the relationship between readmission probability and HbA1c mea primary diagnosis. It demonstrates the value of structured clinical and administrative data for readmission research. It also highlights the importance o and accounting for repeated encounters belonging to the same patient.

2. Literature review – Publication 2

	Verified information
	Predicting 30-Day Hospital Readmission in Patients With Diabetes Using Machine Learning on Electronic Health Record Data.
	Emi-Johnson, O.G. and Nkrumah, K.J. (2025) 'Predicting 30-day hospital readmission in patients with diabetes using machine learning on electronic h Cureus, 17(4), e82437. https://doi.org/10.7759/cureus.82437.
	https://pmc.ncbi.nlm.nih.gov/articles/PMC12085305/
	UCI Diabetes 130-US Hospitals dataset: 101,766 de-identified inpatient encounters from 130 US hospitals, covering 1999–2008.
	Logistic Regression, Random Forest, XGBoost and Deep Neural Networks. SHAP was used to examine influential predictors.
	The authors report feature imputation, scaling and one-hot encoding. They used an 80/20 training/testing split and evaluated accuracy, precision, reca AUC.
	XGBoost achieved the highest reported ROC-AUC (0.667), followed by Logistic Regression (0.642) and Random Forest (0.630). The Deep Neural Ne positive-class recall (0.143), but lower precision (0.186) and ROC-AUC (0.579). SHAP analysis identified previous admissions, number of medications medication-related indicators among influential predictors. The results illustrate why accuracy alone is insufficient for an imbalanced readmission outc

3. Literature review – Publication 3

	Verified information
	The 30-days hospital readmission risk in diabetic patients: predictive modeling with machine learning classifiers.
	Shang, Y., Jiang, K., Wang, L., Zhang, Z., Zhou, S., Liu, Y., Dong, J. and Wu, H. (2021) 'The 30-days hospital readmission risk in diabetic patients: pr machine learning classifiers', BMC Medical Informatics and Decision Making, 21(Suppl 2), Article 57. https://doi.org/10.1186/s12911-021-01423-y.
	https://pmc.ncbi.nlm.nih.gov/articles/PMC8323261/
	The UCI/Health Facts diabetes inpatient dataset derived from clinical records across 130 US hospitals for 1999–2008. The paper reports 101,766 orig 100,244 records in its modelling dataset after preprocessing.
	Random Forest, Naive Bayes and Tree Ensemble classifiers, implemented using the KNIME analytics platform.
	The authors removed variables with extensive missingness, including weight, and removed payer code and medical specialty. Missing race values we mode; records missing any of the three diagnosis attributes were removed. Diagnosis codes were grouped into 16 disease categories. The study com and down-sampling approaches to address class imbalance and used an 80/20 training/testing split.
	For 30-day readmission, Random Forest achieved mean AUC values of 0.640 with over-sampling and 0.661 with down-sampling, outperforming Naiv Ensemble in the reported comparison. Important predictors included previous admissions, age, diagnosis, emergency visits and sex. The paper suppo Random Forest and shows that sampling strategy can affect model performance.

4. Public dataset description

	Verified information
	Diabetes 130-US Hospitals for Years 1999–2008.
	Clore, J., Cios, K., DeShazo, J. and Strack, B. (2014) Diabetes 130-US Hospitals for Years 1999–2008 [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5230J.
	https://archive.ics.uci.edu/dataset/296/diabetes+130-us-hospitals+for-years+1999-2008
	101,766 hospital encounter records.
	47 features, according to UCI metadata. The supplied CSV contains 50 columns: 47 features, two identifier columns (encounter_id and patient_nbr) a column (readmitted).
	The target variable is readmitted, with three values: <30 (readmitted within 30 days), >30 (readmitted after 30 days) and NO (no recorded readmissio
n	For this project, the target is transformed into a binary outcome: 1 = <30; 0 = >30 or NO. This operational definition follows the early-readmission fram dataset paper. It should be stated consistently in the code and report.
supplied CSV	<30 = 11,357 records (11.16%); >30 = 35,545 records (34.93%); NO = 54,864 records (53.91%). After binary transformation: positive class = 11,357 class = 90,409 (88.84%).
	Clinical care data from 130 US hospitals and integrated delivery networks, covering 1999–2008.
	Multivariate data with categorical and integer features.

s	Health and Medicine; associated tasks listed by UCI are classification and clustering.
esented	Patient demographics; admission and discharge information; length of stay; diagnoses; procedures; laboratory testing; medications; and prior inpatien emergency utilisation.
	Yes. Missingness is substantial in some variables. In the supplied CSV, missing values are represented in several fields by the character '?', so these consistent missing-value representation before analysis.
onsideration	The file contains 101,766 encounters but 71,518 unique patients. A patient may therefore contribute more than one encounter. The modelling design level separation or another justified strategy to reduce information leakage between training and test data.
	Creative Commons Attribution 4.0 International (CC BY 4.0), as stated by UCI. Appropriate dataset attribution is required.

5. References

Clore, J., Cios, K., DeShazo, J. and Strack, B. (2014) Diabetes 130-US Hospitals for Years 1999–2008 [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5230J.

Emi-Johnson, O.G. and Nkrumah, K.J. (2025) 'Predicting 30-day hospital readmission in patients with diabetes using machine learning on electronic health record data', Cureus, 17(4), e82437. https://doi.org/10.7759/cureus.82437.

Shang, Y., Jiang, K., Wang, L., Zhang, Z., Zhou, S., Liu, Y., Dong, J. and Wu, H. (2021) 'The 30-days hospital readmission risk in diabetic patients: predictive modeling with machine learning classifiers', BMC Medical Informatics and Decision Making, 21(Suppl 2), Article 57. https://doi.org/10.1186/s12911-021-01423-y.

Strack, B., DeShazo, J.P., Gennings, C., Olmo, J.L., Ventura, S., Cios, K.J. and Clore, J.N. (2014) 'Impact of HbA1c measurement on hospital readmission rates: Analysis of 70,000 clinical database patient records', BioMed Research International, 2014, Article ID 781670. https://doi.org/10.1155/2014/781670.

Scope and interpretation: This document summarises the methods and findings reported by the cited publications. Published performance values are study-specific and should not be treated as expected results for the current implementation. Differences in data cleaning, feature selection, sampling, split strategy and evaluation methods can produce different results.